School of Computing & IT
Web Programming (CIT2011)
Week 1 – Exploring the Web with Developer Tools

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1. Lab Overview

Modern web developers need to understand what happens inside the browser when a webpage loads and when a user interacts with it. In this lab, students use Chrome, or Firefox Developer Tools to inspect HTML, and CSS.

Register to https://onecompiler.com/. You will use this online compiler to share your ongoing labs with your tutor for review.

2. Learning Outcomes

1. Open and navigate browser Developer Tools.
2. Identify HTML elements using the Elements/Inspector panel.
3. Modify HTML and CSS temporarily using Developer Tools.
4. Use responsive/device emulation to test different screen sizes.
5. Use Developer Tools as part of a modern web-development workflow.

3. Software Requirements Students need the following:

- Google Chrome / Mozilla Firefox
- Internet connection
- A text editor such as VS Code

4. Developer Tools Overview

Chrome	Firefox	Purpose
Elements	Inspector	Examine HTML and CSS
Console	Console	Execute JavaScript and view errors
Sources	Debugger	Examine JavaScript/source files
Network	Network	Examine HTTP requests
Application	Storage	Examine cookies/storage
Lighthouse	Lighthouse	Performance/accessibility analysis
Device Toolbar	Responsive Design Mode	Test different screen sizes

PART A Opening Developer Tools

Activity 1: Open Developer Tools

a. Open https://www.w3schools.com/html/default.asp or another publicly accessible website of your choice).
b. Open Developer Tools using right-click → Inspect, F12, Ctrl+Shift+I on Windows, or Cmd+Option+I on macOS.

Question 1. What is the name of the panel used to inspect the webpage's HTML? Elements

Question 2. List five panels available in Developer Tools.

1. Performance 2. Memory 3. Application 4. Security 5. Lighthouse

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PART B — Inspecting HTML

Activity 2: Examine the HTML Structure

Select the Elements/Inspector panel. Locate <html>, <head>, <body>, headings, paragraphs, links, images and navigation elements.

Question 3. Identify one heading element. Record the HTML tag and the text inside it.

HTML tag: <h1> Text: <HTML Tutorial>

Question 4. Find one hyperlink. Record the HTML element and its href value.

HTML element: <a> href= "https://www.w3schools.com"

Activity 3: Modify HTML

Find a visible heading. Right-click the element and select Edit as HTML (or double-click it). Temporarily change it to:

<h1>CIT2011 Developer Tools Lab</h1>

Question 5. What happened to the webpage?

The main heading changed to “CIT2011 Developer Tools Lab” after pasting the text as instructed

and tapping inside the Elements box.

Question 6. Was the original website permanently changed? Explain why.

No. I would not have changed it permanently since this was a temporary test. When reloaded, the page returned with its original title “HTML Tutorial”.

PART C Inspecting CSS

Activity 4: Examine CSS

Select an HTML element containing visible text. Examine the Styles panel and identify CSS properties controlling font size, font family, margin, padding, text colour and background colour. [ELEMENT USED: TITLE]

Property	**Value**
font-size	30px
font-family	“Segoe UI", Arial, sans-serif
margin	10px 0
padding	20px
color	Black #000 Blue
background-color	White

Activity 5: Modify CSS

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Add the following declaration to a heading in the Styles panel:

font-size: 50px; Then

change it to: font-

size: 20px;

Question 1. What happened when the font-size value changed?

When added as 50px then changed to 20px, the heading adjusted in size by becoming smaller.

Question 2p. Add text-transform: uppercase; What happened?

I noticed that the heading changed once I started to type the "up" from the prompt of 'text-transform: uppercase'. After completing my command, the heading changed from HTML Tutorial to HTML TUTORIAL.

Question 3. Change margin and padding. Explain the difference between margin and padding.

Margin is the space outside of an element's border, while Padding is the space between the content and its border. When changing the margin, it affects the space around the element, while changing the padding affects the space inside that element.

PART D — Understanding the Box Model
Activity 6: Inspect the Box Model
Select an element and locate the Box Model diagram.

Question 4. Identify the four major components of the CSS box model.

1. Content 2. Padding 3. Border 4. Margin

PART I Responsive Web Design
Activity 15: Device Emulation
Open Device Toolbar / Responsive Design Mode and test mobile, tablet, laptop and desktop sizes such as 375×667, 768×1024 and 1366×768.

Question 5. What happened to the webpage when the screen width became smaller? When

adjusted with 375*667, the screen's width became smaller and the webpage itself, having less of a

horizontal space.

Question 6. Did the navigation change? Describe what happened.

No, the navigation menu was still visible. Only its spacing was affected by viewport width.

Question 7. Did images resize? Explain your observation.

Yes, the icons adjusted as the screen's width was decreased. This made the images remain in tack of the viewport and made the webpage easier to view on smaller screens.

Activity 16: Test Your Own Responsive Design
Create a file named responsive.html using the following code:

```
<!DOCTYPE html>
<html>
<head>
   <title>Responsive Test</title>
   <style>        body { font-family: Arial;
margin: 20px; }
       .container { display: flex; gap: 20px; }
       .box { padding: 30px; background-color: lightblue; flex: 1; }
       @media (max-width: 600px) {
          .container { flex-direction: column; }
       }
   </style>
</head>
<body>
   <h1>Responsive Web Design</h1>
   <div class="container">
       <div class="box">Box 1</div>
       <div class="box">Box 2</div>
       <div class="box">Box 3</div>
   </div>
</body> </html>
```

Use Developer Tools to test 1200px, 900px, 700px, 600px, 500px and 375px.

Question 8. At approximately what width does the layout change from rows to columns?

At 1200px.

Question 9. Which CSS feature caused the layout to change?

The feature of @media (max-width: 600px) caused the layout to change once edited to 1200px.

Question 10. Complete the table above.

Screen width	Layout	Appearance
1200px	Column (horizontal)	The boxes were stacked under each other
900px	Row (vertical)	All boxes appeared side by side
700px	Row (vertical)	All boxes appeared side by side
600px	Row (vertical)	All boxes appeared side by side
500px	Row (vertical)	All boxes appeared side by side
375px	Row (vertical)	All boxes appeared side by side

FINAL CHALLENGE — Activity 20: Become the Web Detective
A student reports that a webpage is displaying incorrectly on their mobile phone, although it works correctly on a laptop. Use Developer Tools to investigate responsive layout, CSS rules, media queries, images, JavaScript errors, network requests and HTTP status codes.

Question 11. Identify three possible reasons why a webpage may work correctly on desktop but incorrectly on mobile.

1. CSS rules may prevent certain features from working on mobile devices.

2. The images may be too large and would cause horizontal scrolling on smaller devices.

3. The device could possibly not have the responsive media queries.

FINAL REFLECTION Question 12. How does Developer Tools help a web developer understand a webpage?

It allows for the developer to inspect the webpage's HTML, test the difference in screen size and monitor network requests. It also helps the developer to identify problems and test corrections without permanently changing the original webpage.

Question 13. Why is it useful to inspect HTML and CSS rather than simply looking at the webpage?

It is useful because it helps the developer to understand how the webpage is structured and to

identify incorrect tags, missing elements, and layout issues.

Question 14. Why is the Network panel important to a web developer?

It is important because it shows the resources a webpage tools, including CSS files and other

requests. It helps the developer to identify failed requests and HTTP status codes such as 404 errors.

Question 15. Why is responsive testing important?

Because it ensures that a webpage displays correctly on different devices which improves readers' readability and accessibility.

Question 16. How can Developer Tools help identify bugs?

It can be identified through the panels such as Elements, Console, Styles and Network where developers can view error messages, test CSS changes and identify failed requests in order to fix problems.

Question 17. Explain how HTML, CSS and JavaScript work together to create an interactive webpage.

HTML provides structure and contents such as headings, images and links; CSS controls the appearance and layout

JavaScript adds interactivity such as responding to button clicks, and load time. They work together to create

functionality of the webpage.

Question 18 – Build a webpage
Use the template provided by your tutor to build a webpage using HTML5 and basic CSS3. Webpage must include, basic HTML5 tags such as, image, paragraph, navigation, heading, sections, footer, links, etc.

```
<!DOCTYPE html>

<html lang="en">

<head>

   <meta charset="UTF-8">

   <meta name="viewport" content="width=device-width, initial-scale=1.0">

   <title>Sweet Sweet Jamaica</title>
```

```
<style>
    body{
        font-family: Times New Roman, sans-serif;
        margin: 20px;
        background-color: green;
    }

    header {
        background-color: gold;
        color: white;
        text-align: center;
        padding: 25px;
    }

    nav {
        background-color: lightgreen;
        padding: 15px;
        text-align: center;
    }

    nav a {
        color: white;
        text-decoration: none;
        margin: 15px;
```

```
}

section {
    background-color: white;
    margin: 20px;
    padding: 20px;
    border-radius: 8px;
}

img {
    width: 375px;
    max-width: 100%;
    border-radius: 8px;
}

footer {
    background-color: black;
    color: white;
    text-align: center;
    padding: 12px;
}

@media (max-width: 600px) {
    nav a {
        display: block;
```

```
            margin: 10px;
        }
    }
  </style>
</head>

<body>

  <header>
    <h1>Sweet Sweet Jamaica</h1>
    <p>Out of Many, One People</p>
  </header>

  <nav>
    <a href="#about">About Jamaica</a>
        <a href="#culture">Our Culture</a>
    <a href="#places">Places to Visit</a>
  </nav>

  <main>
    <section id="about">
      <h2>About Jamaica</h2>
      <p>Jamaica, otherwise known as the 'capital of the Caribeean', is a beautiful Caribbean island known for
        its beaches, friendly people, amazing food and
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        rich cultural heritage.</p>
      <img src="https://placehold.co/375x180?text=Beautiful+Jamaica"
        alt="Beautiful Jamaica">
    </section>

    <section id="culture">
      <h2>Jamaican Culture</h2>
      <p>Jamaican culture is known for the rich reggae music, jerk
        cuisine, athletics and vibrant traditions.
        Jamaica is a country full of talent and creativity. "We likkle but we                tallawah"</p>
    </section>

    <section id="places">
      <h2>Places to Visit</h2>
      <p>Jamaica has many wonderful attractions, including
        Dunn's River Falls, the Blue Mountains and Negril's
        famous Seven Mile Beach.</p>
      <a href="https://www.visitjamaica.com/">
        Explore Jamaica
      </a>
    </section>


  </main>
```

```
  <footer>

    <p>&copy; 2026 Sweet Sweet Jamaica | Learn About Our Island</p>

  </footer>



</body>

</html>
```

Practical Deliverables

- Completed lab sheet (Questions 1–18).
- At least six screenshots: HTML inspection, CSS inspection
- responsive.html